Sp/B/ark/ade

Bitcoin L2s without a soft fork: what they promise, what you trust

@bonomat

BushBash Palm Cove · July 2026

Ark



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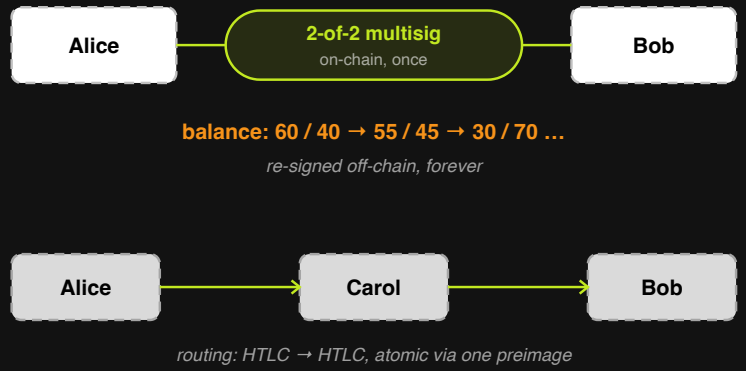
Lightning

Bitcoin's first L2, and where it hurts

Lightning in 60 seconds

- On-chain Bitcoin: slow, fees spikes, not private.
- Lightning's answer: a **payment channel**: one on-chain **2-of-2 multisig** between two parties.
- Inside it, they re-sign the balance split **off-chain**, as often as they like. Only open + close touch the chain.
- No direct channel? **Route** through others; HTLCs make every hop atomic.

–> Result: instant, cheap, high-volume payments. It works™ : exchanges, most of retail BTC payments today.



Why Not Lightning

- 💸 **A channel per user**: Onboarding the next billion = billions of channel opens.
- 📩 **Inbound liquidity**: You can't *receive* until someone locks coins toward you. New users must buy or beg for inbound capacity.
- 🔌 **Always online**: receiving needs your node up. Miss a revocation, and a stale-state close can cost you. Hence watchtowers.
- ⚠️ **Force-close pain**: unhappy path = on-chain tx at the worst time, timelocked funds, fee spikes.
- 🧭 **Routing failures**: larger payments might fail to find a path.

In practice most users ended up on **custodial** Lightning wallets, the very thing we were trying to avoid.

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The New L2s

Shared UTXOs to solve everything™

The promise

- Instead of one tx/*2 users*, one tx/**n users**.
- ⚡ instant, cheap payments, **no inbound liquidity problem**
- 📱 **receive while offline**, no node to run
- 🧾 no per-user on-chain footprint to get paid
- 🔑 still **self-custodial**: exit without permission

–> Spark, Bark and Arkade are three different implementations:

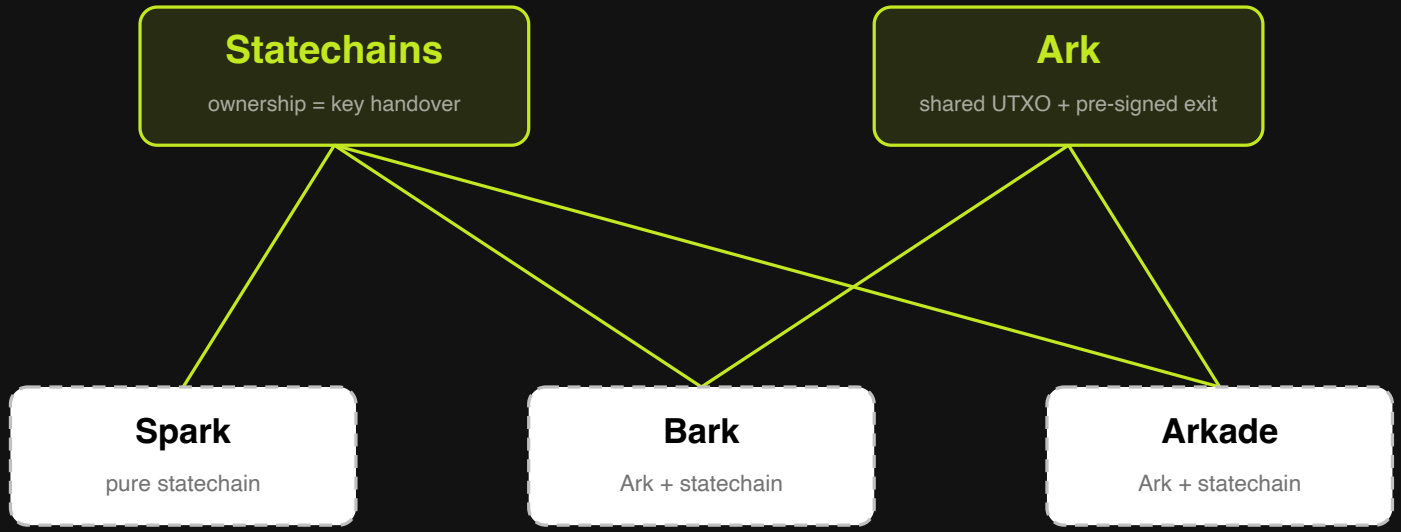
> How do they work and differ?

The axes we'll compare on:

- ⚙️ **Technical implementation**: how does it work under the hood?
- 🔐 **Trust model**: what exactly must be honest?
- 🚪 **Exit**: how do you leave without permission?
- ⏰ **Liveness**: what happens if you go offline?
- 🛠️ **Programmability**: can you build on it?

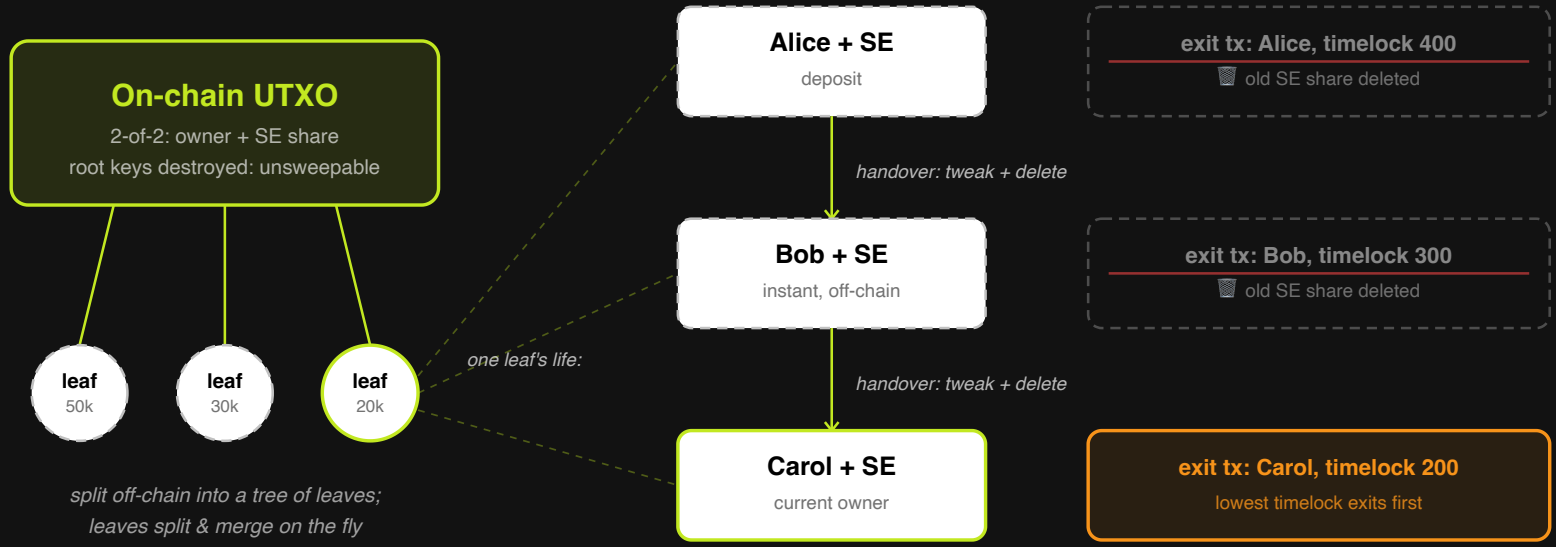
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Technical Comparison



Bark and Arkade: Ark rounds for settlement, statechain-style handover for instant sends

Statechains: how Spark moves coins



the coin never moves; who can sign for it moves, and the newest owner always wins the exit race

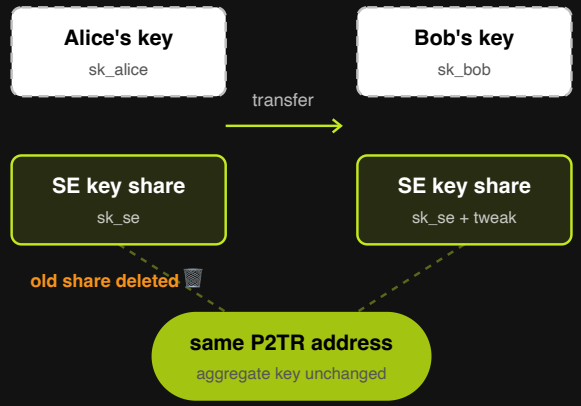
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Spark

Statechains: "we deleted the old key, trust us"

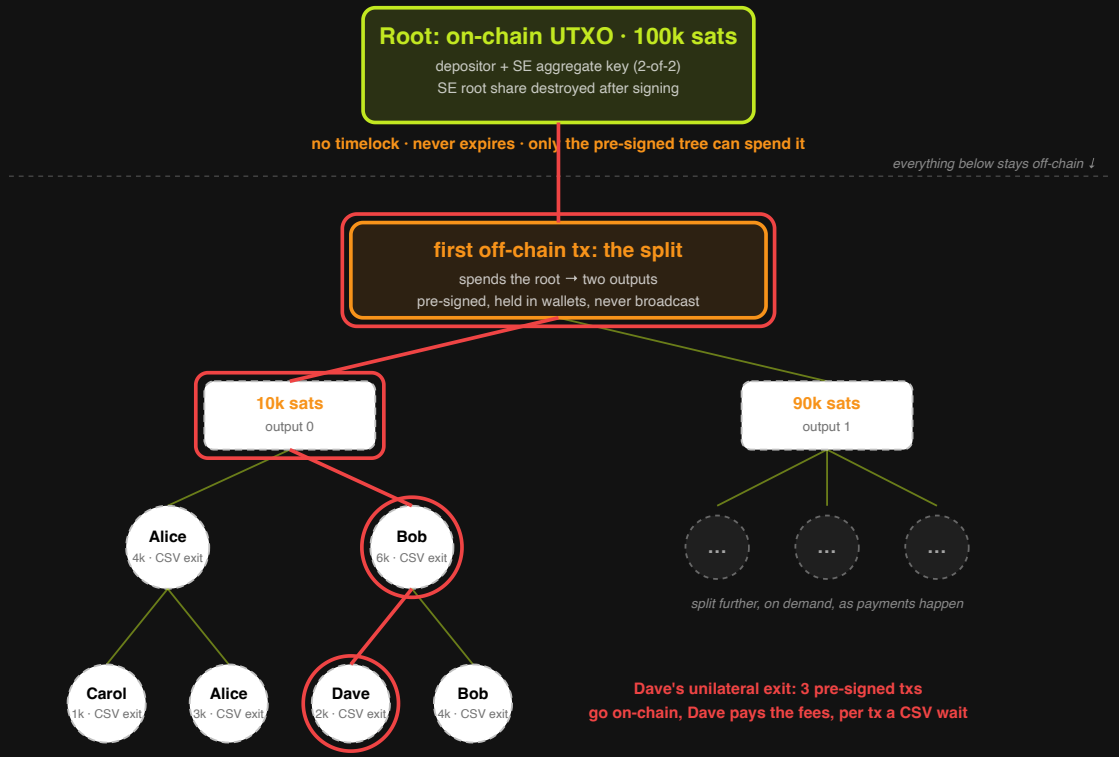
Spark: key handover

- Built by **Lightspark** on **statechains**.
- **Operator federation** (Lightspark, Flashnet, Breez) holds one key share, you hold the other.
- **Your signature is always required.**
- Transfer: SE **tweaks its share**, **deletes the old one**, signs a new exit tx with a **lower timelock**.
- No on-chain tx. **Offline receive.**
- **Relative timelocks**: to unilaterally exit.



The coin never moves. **Who can sign for it** moves.

The Spark tree: trust the depositor



Spark: Summary

Pros

- Instant, free Spark→Spark transfers, **no liquidity provisioning**, no rounds, no expiry.
- Offline receive; no on-chain footprint to get paid.
- Lightning via SSPs: 0.25% + routing out, 0.15% in. No channels for the user.
- Unilateral exit: yes, chain of txs
- **Stablecoin**: BTKN tokens (**USDT** and **USDB**), implemented by: WoS, Blitz, Breez, B

Cons

- Security = SE **actually deleted** the old key.
- Federation is 3 companies. Liveness of the federation required for every transfer.
- No scripts, no contracts. Multisig only.
- Closed source spark core (btc custody, seed storage, lightning swaps)
- **Where does the liquidity come from?**
- **Who is facilitating lightning payments?**

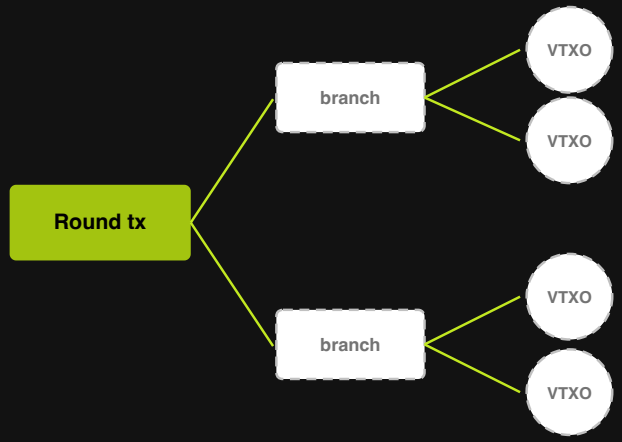
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Ark

The original protocol: shared UTXOs, rounds, and trees

Ark: rounds and trees

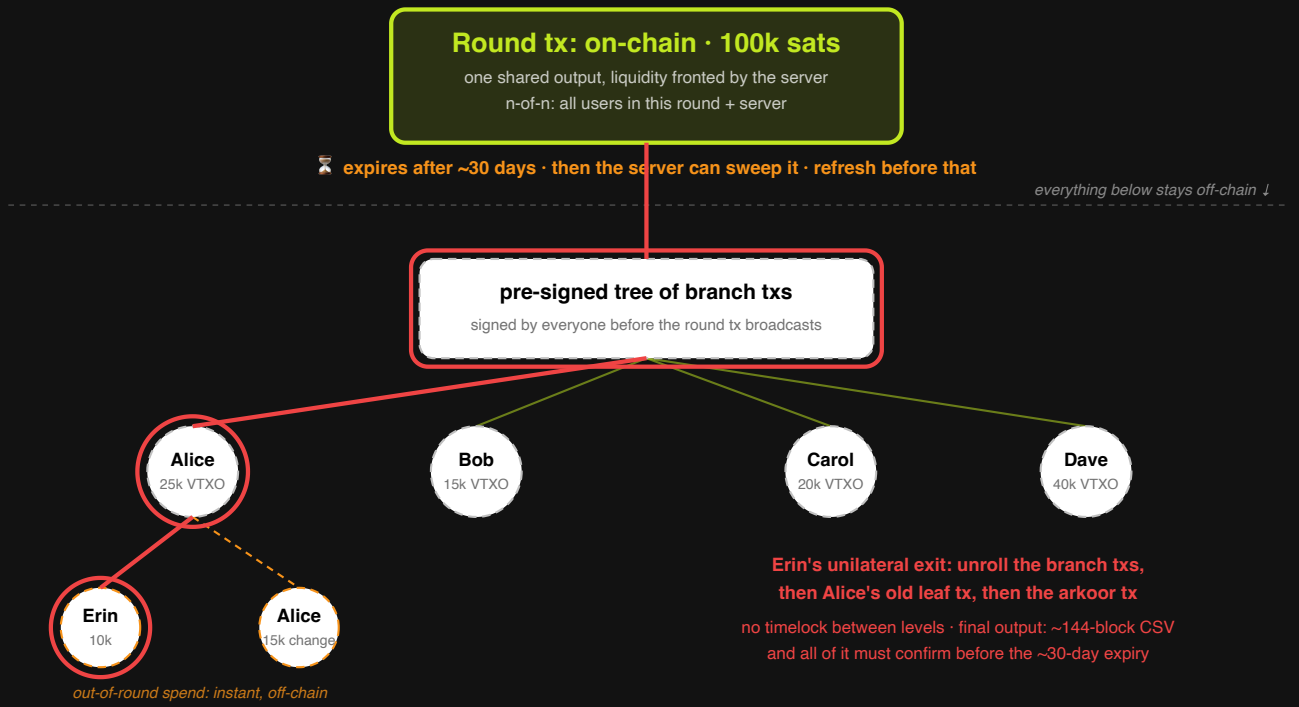
- Proposed by **Burak (2023)**: one UTXO among many users, renew it in **rounds**.
- The **Ark server** periodically broadcasts **round tx** that funds a **shared output**, fronting the liquidity.
- This output splits into a pre-signed tree (VTXOs).
- Every tree node: cooperative **n-of-n** (everyone below + server) or **timelocked recovery**.
- VTXOs **expire**: after the round's lifetime the server **sweeps** the shared output to recover its liquidity; users * *refresh** into a new round before that.



every edge = a pre-signed tx in your pocket

Unilateral exit = broadcast your branch, wait out a CSV delay, leave. Works until expiry.

The Ark round tree



same virtual-tree trick as Spark, but: the anchor expires, and the whole tree is signed by everyone up front

Bark: Ark + Statechains

- **Second** (Steven, Erik, Neil, ex-Blockstream).
- First public Ark mainnet: **June 2026**.
- No covenants needed: each round's tree is **signed interactively** by all participants; be online or **delegate** your refresh.
- Rounds roughly **hourly**, VTXO lifetime **~30 days** (both server-configurable); quad trees instead of binary trees, keep exits shallow.
- Payments = **arkoor** (*"Ark out-of-round"*): instant, server co-signs, statechain-flavored until refreshed.
- In production: Arkee, Bark, Noah

Bark: Summary

Pros

- Board/refresh VTXOs are **trustless**: real unilateral exit, pure Bitcoin txs.
- Instant **arkoor** payments; offline receive.
- **Custom finality**: join a round anytime; the refresh removes all trust in former senders.
- Lightning both ways via server as gateway (HTLC-on-VTXO), no channels.
- Open protocol, **self-hostable server** (`captaind`), Rust, full SDK.

Cons

- VTXOs **expire (~30 days)**: Liquidity cost each refresh (~0.07%).
- Refresh needs **online during a round** (can be delegated).
- **Arkoor** payments: receiver trusts *sender + server don't collude*.
- Server is also the **only Lightning/swap provider**: every LN payment routes through it; LN receive trusts a server key deletion.

Focused on fast and easy payments.

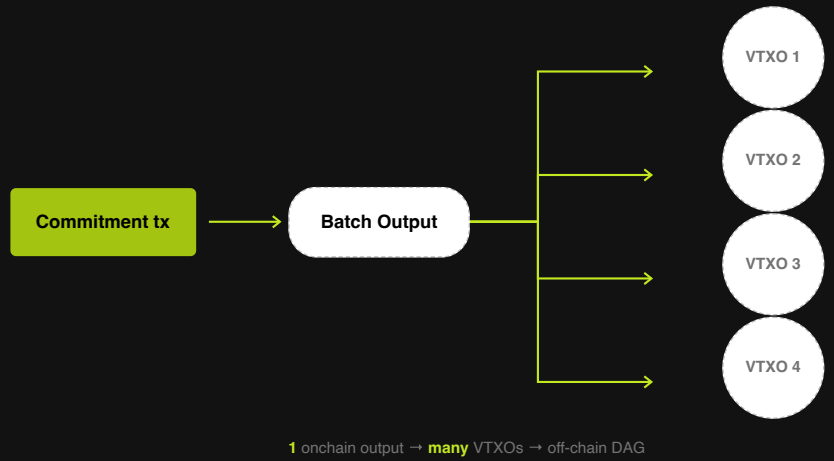
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Arkade

Ark + a programmable execution layer

Arkade: Ark + Statechain + extended Scripts

- Built by **Ark Labs**. Mainnet quietly since **Aug 2025** (Baltic Honeybadger), public beta **Oct 2025**.
- Same Ark base as Bark: **VTXOs** in batch outputs, commitment txs, forfeits, expiry + renewal, unilateral exit.
- **Arkade Script**: Bitcoin Script + introspection opcodes → **covenants**: vaults, swaps, channels.
- **Arkade Assets**: issued assets living on VTXOs; **USDT** coming (Tether-backed).



Arkade: Summary

Pros

- Everything Ark gives: instant preconfirmed payments, offline receive, unilateral exit.
- **Custom finality**: settle into a batch anytime; removes all trust in former senders.
- **Covenants today**: Arkade Script sees the spending tx; vaults, non-interactive swaps, channels *inside* VTXOs.
- **Assets on VTXOs**, same exit guarantees; USDT coming.
- Lightning via **independent swap providers**: Satora, Boltz.

Cons

- **Single operator** (Ark Labs); preconfirmations trust it not to co-sign a conflict.
- Covenants are **not L1-enforced**: they run in a **TEE co-signer** (the Emulator, attested)
- Same Ark **expiry + renewal** burden (delegation helps).

Focused on builders: a programmable execution layer on Bitcoin.

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Summary

The scorecard

	Spark	Bark	Arkade
Design	Statechains, key handover	Ark + statechains	Ark + statechain + scripts
Open source	client/server, spark core closed	**fully open source**	**fully open source**
Operator	Federation: Lightspark, Flashnet, Breez	Single operator	Single operator (Ark Labs) + TEE signer
You trust…	threshold *deleted* old key shares (unverifiable)	trustless once refreshed; arkoor: sender+server no-collusion	trustless once refreshed; sender+server no-collusion
Custom finality	none, deletion trust forever	✅ refresh into a round drops all sender trust	✅ settle into a batch drops all sender trust
Expiry	none, leaves live forever	**~30 days**, refresh in rounds	**~30 days**, refresh in rounds
Unilateral exit	pre-signed chain, decrementing timelocks	unroll tree, ~144-block CSV	unroll tree + chain, 1008-block CSV
Offline receive	✅ fully	✅ preconfirmed	✅ preconfirmed
Lightning	via SSPs, 0.25%+routing out / 0.15% in	server is the LN gateway, 0.2%-0.5%	Satora, Boltz swaps (0.1% routing out / 0.25% in)
Assets	BTKN: **USDT + USDB live**	none	Arkade Assets, USDT planned
Programmability	none	Bitcoin Script only	**introspection covenants**
Mainnet	beta since Apr 2025 · WoS, Theya, Blitz	since Jun 2026 · Noah, Arkee, BTCPay	Aug 2025 / beta Oct 2025 · Satora, Boltz, Bull Bitcoin

Which one is for you?



Resources

Docs

- Spark: docs.spark.money
- Bark: second.tech/docs
- Arkade: docs.arkadeos.com

Try it

- Spark wallets:
  - Wallet of Satoshi: walletofsatoshi.com
  - Blitz: blitz-wallet.com
  - Breez SDK: sdk.breez.technology
- Bark wallets:
  - Noah: github.com/smolcars/noah
  - Arkee: arke.cash
- Arkade wallet: arkade.space

Questions?

Spark: docs.spark.money · Bark: second.tech/docs · Arkade: docs.arkadeos.com

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